

Extensions of Count-Min Sketch

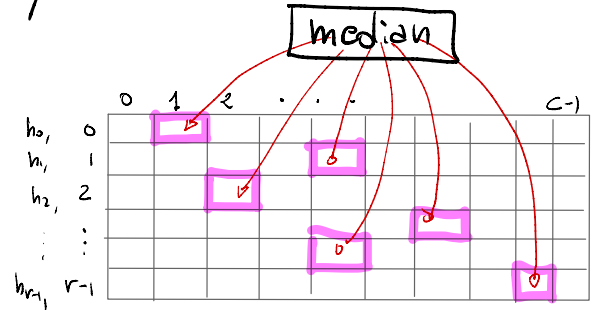
- ① What if we can also decrease? That is, update $F[i]$ -- is supported (actually this holds for any value, not necessarily ± 1)

Still, we have $\tilde{F}[i] = F[i] + X_{ji}$

≤ 0

As we want to choose j s.t. X_{ji} is sufficiently close to the ideal value of zero, we define

$$\tilde{F}[i] = \text{median} \{ T[0][h_0(i)], \dots, T[r-1][h_{r-1}(i)] \}$$



$$\blacktriangleright \underbrace{F[i] - 2\varepsilon \|F\|} \leq \tilde{F}[i] \leq \underbrace{F[i] + 2\varepsilon \|F\|} \Leftrightarrow \underbrace{|\tilde{F}[i] - F[i]|}_{\text{garbage } |X_{ji}|} \leq 2\varepsilon \|F\|$$

Here we follow a boosting approach based on CB (Chernoff's Bounds)

$$Y_j = \begin{cases} 1 & \text{if } |X_{ji}| \leq 2\varepsilon \|F\| \\ 0 & \text{o.w. (BAD)} \end{cases} \quad \text{with probability } p = 1 - \frac{1}{2e} > \frac{3}{4}$$

(recall that we need) $\underbrace{p}_{p > \frac{1}{2}} > \frac{3}{4}$

$$\begin{aligned} \Pr[Y_j=1] &= \Pr[|X_{ji}| \leq 2\varepsilon \|F\|] = 1 - \Pr[|X_{ji}| > 2\varepsilon \|F\|] \stackrel{\text{H}}{\geq} \dots \\ &> 1 - \frac{E[|X_{ji}|]}{\underbrace{2\varepsilon \|F\|}} \stackrel{(*)}{=} 1 - \frac{\frac{\varepsilon}{e} \|F\|}{\cancel{2\varepsilon \|F\|}} = 1 - \frac{1}{2e} \end{aligned}$$



$$Y = \sum_{j=0}^{r-1} Y_j = \# \text{ successful rows}, \quad \mu = E[Y] = rp > \frac{3}{4}r$$

As in Morris' approximate counting,

$$\tilde{F}[i] \text{ is bad} \Rightarrow Y \leq \frac{r}{2} \Rightarrow \dots \Rightarrow |Y - \mu| \geq \mu - \frac{r}{2} > \frac{3}{4}r - \frac{r}{2} = \frac{1}{4}r$$

median $\nearrow Y > \frac{r}{2}$ is on for the median!

CB come into play: $\Pr[|Y - \mu| > \underbrace{\lambda\mu}_{\frac{r}{4}}] < 2 \cdot e^{-\frac{\lambda^2\mu}{3}}$

setting $\lambda\mu = \frac{r}{4} \Rightarrow \lambda = \frac{r}{4\mu} < 1$

$$\Rightarrow e^{-\frac{\lambda^2\mu}{3}} < e^{-\frac{r}{4 \cdot 3}} = e^{-\frac{r}{48}} < \delta$$

Hence $r = \Theta(\ln \frac{1}{\delta})$ as we did in the CM-sketch



② Interval queries. $[x..y]$

$$0 \leq x \leq y \leq n-1$$

range of items i s.t. $x \leq i \leq y$

(e.g. a domain of consecutive IP addresses)

$$F_{xy} = \sum_{x \leq i \leq y} F[i] \approx \tilde{F}_{xy}, \epsilon, \delta$$

BASELINE

$$\tilde{F}_{xy} = \sum_{x \leq i \leq y} \tilde{F}[i] \text{ has two issues}$$

→ query time is $O((y-x+1) \ln S^{-1})$
and can be VERY large

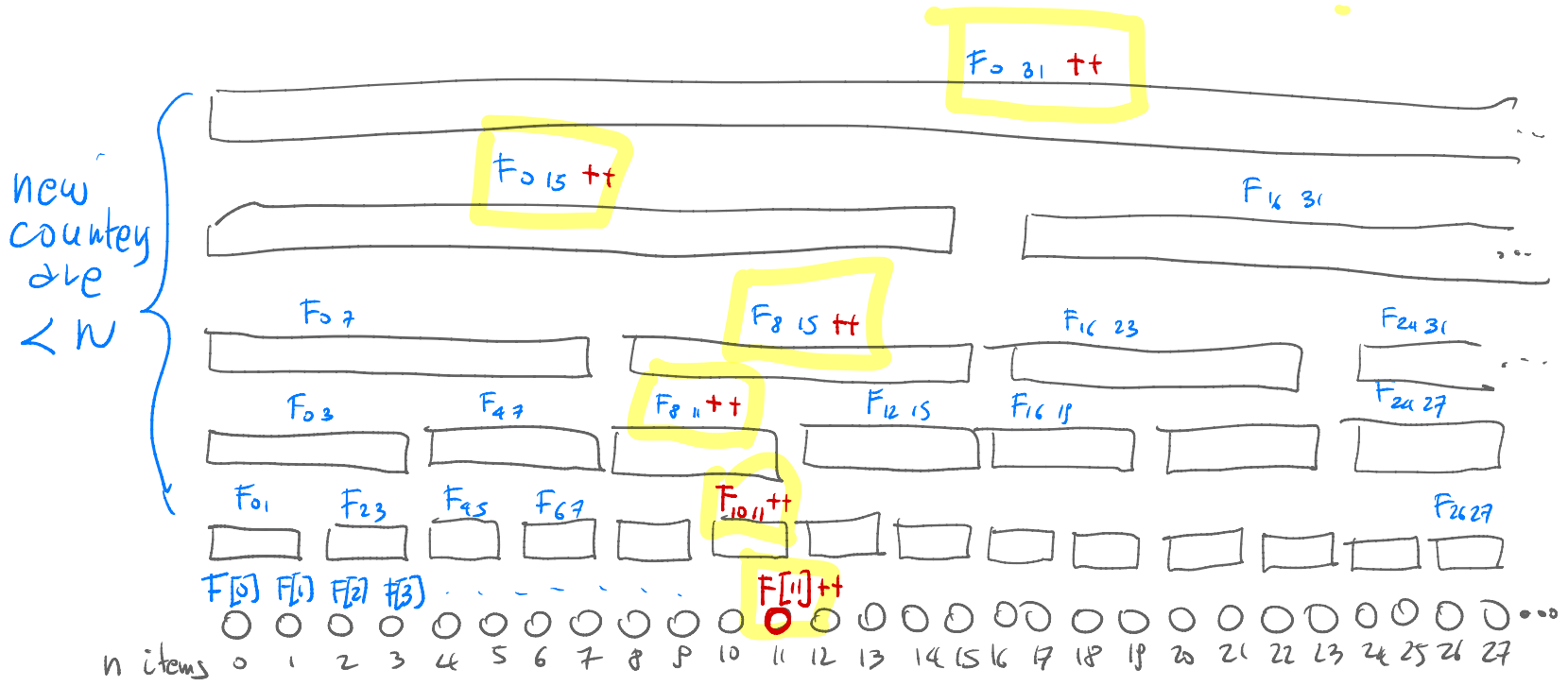
→ the error can be VERY large and out of control

BEST APPROACH : USE "DYADIC" INTERVALS

their length
IS a power of 2

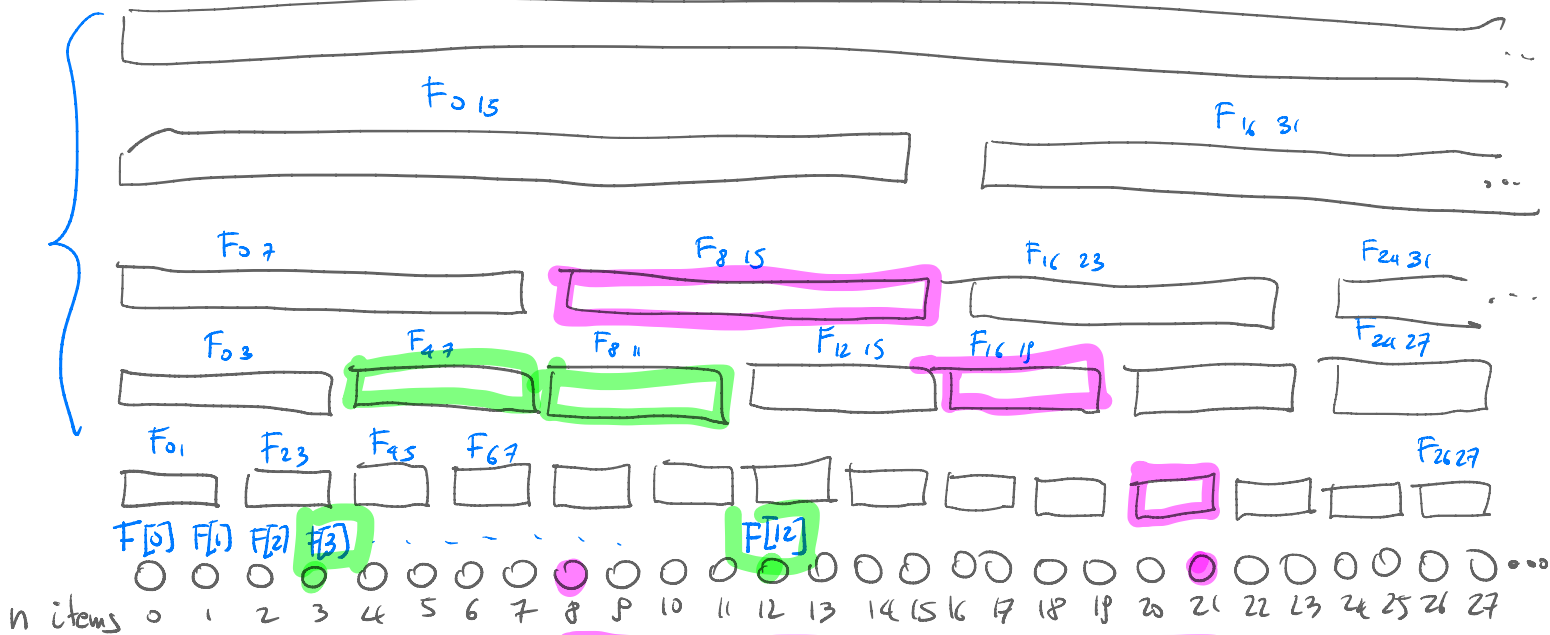
We "keep" $< 2n$ counters instead of n counters

- When e.g. $F[i]++$, it is like incrementing the counters along a leaf-to-root path in a complete full (implicit) binary tree (e.g. see $F[11]++$)
cost is $O(\lg n \cdot \lg S)$ time



How to query F_{xy} $\rightarrow$ sum of $\leq 2 \lg n$ counters

example with $x=3, y=12 \Rightarrow F_{3,12} = F[3] + F_{4,7} + F_{8,11} + F[12]$

$$X=8, Y=21 \Rightarrow F_{8,21} = F_{8,15} + F_{16,18} + F_{20,21}$$


obs on each level at most two dyadic intervals are taken
proof if there were three or more, at least three of them are consecutive, and so the parent should be taken

Let's write $F_{xy} = \sum_{a,b} F_{ab}$
 $\leq 2 \lg n$ addends

$$\Rightarrow \tilde{F}_{xy} = \sum_{a,b} \tilde{F}_{ab}$$

$O(\lg n \lg S^{-1})$
time

As for the error: $F_{xy} \leq \tilde{F}_{xy} \leq F_{xy} + 2 \cdot \lg n \cdot \varepsilon \|F\|$

We use a little trick: "cumulative" garbage $\tilde{F}_{xy} = F_{xy} + X_{xy}$

$E[X_{xy}] \leq 2 \lg n \cdot \frac{\varepsilon}{e} \|F\| \Rightarrow \text{MI}$

$\Pr[X_{xy} > 2 \cdot \lg n \cdot \varepsilon \cdot \|F\|] < \frac{E[X_{xy}]}{2 \cdot \lg n \cdot \varepsilon \cdot \|F\|} \leq \frac{2 \cdot \lg n \cdot \frac{\varepsilon}{e} \cdot \|F\|}{2 \cdot \lg n \cdot \varepsilon \cdot \|F\|} = \frac{1}{e}$

THE REST IS THE SAME ...
 $(\frac{1}{e})^r \leq S$